HS Harz FuSa, Friedrichstraße 57

Airbag ECU

Messung und Verarbeitung der Sensoreingänge zur Auslösung des Airbags
Measurement and Processing of Sensor Signals to Process an Airbag Operation

Phase 2: Hazard Analysis and Risk Assessment (HARA)

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Contents

1 Meetings within the project

Topic	Time	Room	Participants
Item Definition	11:00	Zoom	ALL
Bericht:	*none*		

2 Unresolved topics

No unresolved topics

3 Glossar

Tabelle 1: Glossar

Begriff / Term	Erklaerung / Explanation
ACU	Airbag Control Unit the ECU responsible for crash detection and airbag deployment control
ADC	Analog-to-Digital Converter converts analog sensor voltage to a digital integer value
ARM Cortex-M3	32-bit RISC processor core used in the Arduino Due (SAM3X8E)
ASIL	Automotive Safety Integrity Level A to D, D being highest requirement (ISO 26262)
CAN	Controller Area Network automotive serial communication bus standard (ISO 11898)
ECU	Electronic Control Unit embedded computing module in a vehicle
ESD	Electrostatic Discharge overvoltage event that can damage electronic components
EMI	Electromagnetic Interference unwanted electrical noise affecting circuit operation
GPIO	General Purpose Input/Output configurable digital pin on a microcontrol-ler
HW	Hardware
ISO 26262	International standard for functional safety of road vehicle electri-cal/electronic systems
LDO	Low Drop-Out Regulator linear voltage regulator with low input-output voltage differential
MCU	Microcontroller Unit
MEMS	Micro-Electro-Mechanical System miniature sensor technology used for accelerometers and gyroscopes
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor used as switching element in output drivers
Safing Condition	A secondary, independent confirmation of crash event required before deployment is enabled (dual-point safety architecture)
SWR	Software Requirement functional requirement assigned to the software module
SW	Software
TVS	Transient Voltage Suppressor protection diode for input overvoltage clamping

4 Hazard Analysis and Risk Assessment (HARA)

The HARA process, mandated by ISO 26262-3, drives the identification of potential system malfunctions. By mapping technical failure modes to operational risks, we define the boundaries of safe operation. Based on the IJERT FMEDA and Fraunhofer system-level modeling.

The Serial Peripheral Interface (SPI) is a critical focus. As the SPI Master, the ECU's internal shifter logic and baud rate generator are vital; any corruption here such as a fire command issued due to corrupted noise or a wrong decode of clock selection can lead directly to an unintended deployment.

4.1 Identified Hazards

Tabelle 2 Identified Hazards

No.	Hazard	Technical Cause	Effect
H1	Unintended airbag deployment	Shifter logic corruption or data corruption in the transmit register, where a malformed SPI message is interpreted as a firing command.	Loss of vehicle control and direct high-velocity trauma.
H2	Failure to deploy during a crash	Failures in the baud rate generator or protocol generation logic (SCLK/SS failures) preventing sensor data acquisition.	Total loss of passive safety, resulting in life-threatening or fatal injuries.
H3	Passenger airbag deployment with baby seat present	Failure in the logic-gate processing of the occupancy sensor data.	High risk of infant fatality.
H4	Delayed deployment	Protocol delays or interrupt service failures causing activation outside the FTTI.	Occupant may already be out of position, making the airbag a source of injury rather than protection.
H5	Incorrect airbag selection during activation	Wrong decode of clock selection or interrupt count errors leading to improper actuation logic.	System deploys an inappropriate squib (e.g., right-side curtain for a left-side impact).

4.2 Risk Assessment Methodology: Severity, Exposure, and Controllability

The ISO 26262 framework classifies risk by evaluating the potential harm through the lens of environmental exposure and human-led mitigation.

Tabelle 3 Risk Assessment Methodology

Parameter	Scale	Definition
Severity (S)	S0-S3	Quantifies the intensity of potential harm. For the Airbag ECU, failures are typically S3, indicating life-threatening or fatal injuries.
Exposure (E)	E0-E4	Measures the probability of the vehicle being in a situation where the hazard can occur. E4 represents highly probable scenarios (e.g., high-speed driving).
Controllability (C)	C0-C3	Assesses the driver's ability to prevent injury once a malfunction occurs. Unintended airbag deployment is classified as C3 (uncontrollable), as a driver cannot mitigate a high-velocity deployment while in motion.

Strategic Justification: The combination of S3, E4, and C3 variables mandates an ASIL D classification. This rating represents the highest rigor in automotive safety assurance, requiring the implementation of advanced hardware metrics and rigorous diagnostic coverage as defined in ISO 26262-5.

4.3 ASIL Determination and Classification Matrix

The following table summarizes the ASIL assignments for the Airbag ECU. Assignments are driven by the catastrophic nature of passive safety failures.

Tabelle 4 ASIL Determination and Classification Matrix

Hazard	Severity (S)	Exposure (E)	Controllability (C)	ASIL Level
Unintended Deployment	S3	E4	C3	ASIL D
No Deployment during Crash	S3	E4	C3	ASIL D
Suppression Failure (Baby Seat)	S3	E3	C3	ASIL C
Delayed Deployment	S3	E4	C2	ASIL C
Wrong Airbag Activation	S3	E4	C3	ASIL D

Note: While SG3 and SG4 are rated ASIL C due to lower exposure (frequency of baby seats) or higher controllability (timing delays), the overall ECU design is governed by the ASIL D mandate of the primary firing logic.

5 Figure and table listing

6 List of dependent documents

No.	Dokument name	Version	Date	File name
1	Hazard Analysis and ASIL Determination for Airbag ECU	V02	02.06.2026	Hazard Analysis and ASIL Determination for Airbag ECU V02.pdf
2	Hazard Analysis and ASIL Determination for Airbag ECU	V01	07.05.2026	Hazard Analysis and ASIL Determination for Airbag ECU_V01.pdf
3	ECU Airbag System Architecture Documentation	V02	27.05.2026	ECU Airbag system architecture documentation V2.pdf

7 Literature

[1] M. Skutek, M. Mekhaiel, and G. Wanielik, A precrash system based on radar for automotive applications, in IEEE IV2003 Intelligent Vehicles Symposium, Columbus, OH, USA, 2003, pp. 37-41. doi: 10.1109/IVS.2003.1212879.

[2] H. Komaki, A. Kitabatake, T. Koyama, T. Tabata, and H. Konishi, Denso-Ten Technical Journal. Available: https://www.denso-ten.com/business/technicaljournal/pdf/18-4.pdf

[3] RDAIRBAGPSI5 Airbag Reference Platform, Freescale Semiconductor, Inc. Available: https://community.nxp.com

[4] Automotive power product brief TLE 7714/7734, Infineon Technologies. Available: https://www.infineon.com/dgdl/TLE7714-34_PB_final.pdf

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8 Version history

Document name	date	Change log
Airbag_ECU_Phase_2	02.06.2026	v1.00 Initial version. HARA compiled from Hazard Analysis and ASIL Determination for Airbag ECU V02 (02.06.2026). V02 supersedes V01 (07.05.2026).